Spring 2026 Topic Reviews for Math 54

Faculty:	Profs. Arun Sharma and Vera Serganova
Facilitator:	Jason Ferguson, jason.ferguson@berkeley.edu
Time and Location:	Tu 3:30-5 on Zoom and in 201A Student Center, Th 3:30-5 on Zoom and 6 Evans
Jason's Office Hours:	As of 3/16, M 12-1, TuTh 2-3 See here for the most up-to-date Drop-In Schedule

Practice Final for 3:30-5pm Thursday, May 7, 2026's Math 54 Problem-Solving Session

Objectives: By engaging actively with this session, participants will

1. Assess in real-time their problem-solving skills when solving individual novel, Final Exam Exam-level problems in an exam setting
2. Identify types of problems that would benefit them to practice more

Note: I have not seen Prof. Serganova's OR Prof. Sharma's Math 54 Final yet. I tried to make this Practice Final be representative of what I believe to be the length, question types, and difficulty of the their Finals, but cannot guarantee that that is the case. In particular, I cannot guarantee this Final is comprehensive. In other words the actual Final may ask different types of questions and may cover content not on this Practice Final.

Notice: Prof. Sharma's final has 13 problems (9 free-response, 3 sets of 5 true/false with no explanation, and 1 bonus), while Prof. Serganova's final has 9 problems (2 of which are true/false).

This practice final has 12 problems (8 free-response, 3 sets of 5 true/false, and 1 bonus/harder question), so it does not exactly match the question number or type of problem in either final, but the overall length feels representative.

Also notice: There are two problem 7s and two problem 8s; one intended for Prof. Serganova's students and one intended for Prof. Sharma's students.

Breakdown:

- Free-response: Problems 1-6 (25 pts each)
- Free-response, but different problems for different lectures Problems 7AB-8AB (25 pts each)
- True/False: Problems 9-11 (25 points each)
- Free-response/bonus: Problem 12 (15 points each)

Print your name: ______________________________

SID Number: ______________________________

TIME AND CONDITIONS: 170 minutes; (for Prof. Sharma) one note and formula sheet provided no other notes allowed (for Prof. Serganova) bring a formula sheet you create; closed book/internet; no calculator/computer

Space for notes on test-taking strategies:

* I aimed to make this slightly harder than actual final.

* Computations not always nice.

2, 5, 6, 7B, 8B, T/F.

1. (25pts) Let A be the following matrix:

Standard.

$$\begin{bmatrix} 0 & 1 & 3 & 0 \\ 1 & 0 & 0 & 4 \\ 0 & 0 & 0 & 2 \\ 0 & 2 & 5 & 2 \end{bmatrix}$$

Do the following parts in any order you wish.

(a) Compute $\det A$

(b) Compute A^{-1}.

(a) $-1 \det \begin{bmatrix} 1 & 3 & 0 \\ 0 & 0 & 2 \\ 2 & 5 & 2 \end{bmatrix} = -1(-2) \det \begin{bmatrix} 1 & 3 \\ 2 & 5 \end{bmatrix} = (-1)(-2)(-1) = \boxed{-2}$

(b) reduce $[A \mid I]$ to get $[I \mid B]$: B is A^{-1}.

$$\left[\begin{array}{cccc|cccc} 0 & 1 & 3 & 0 & 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 4 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 2 & 0 & 0 & 1 & 0 \\ 0 & 2 & 5 & 2 & 0 & 0 & 0 & 1 \end{array}\right] \longrightarrow \left[\begin{array}{cccc|cccc} 1 & 0 & 0 & 0 & 0 & 1 & -2 & 0 \\ 0 & 1 & 0 & 0 & -5 & 0 & -3 & 3 \\ 0 & 0 & 1 & 0 & 2 & 0 & 1 & -1 \\ 0 & 0 & 0 & 2 & 0 & 0 & 1 & 0 \end{array}\right]$$

I got to this without any fractions. (Steps skipped)

Answer:

$$\boxed{\begin{bmatrix} 0 & 1 & -2 & 0 \\ -5 & 0 & -3 & 3 \\ 2 & 0 & 1 & -1 \\ 0 & 0 & \frac{1}{2} & 0 \end{bmatrix}}$$

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3:49

2. (25pts) Let V be the vector space of all twice-differentiable functions on $(-\infty, \infty)$. Let W be the vector space of all functions on $(-\infty, \infty)$. Let $T\colon V \to W$ be the function:

$$T(y) = y'' + 4y' + 5y,$$

   (a) Find a basis for the kernel of T.

   (b) Find all solutions y to the following equation using undetermined coefficients:

$$T(y) = e^{-2t} + \sin(t)$$

(a)/(b) Asking a very standard problem in a weird way.

How to start: "Kernel of T" means "All solutions y to $T(y) = \vec{0}$". (in general). For this: kernel is all solutions to

$$y'' + 4y' + 5y = 0$$

$$r^2 + 4r + 5 = 0$$
$$r^2 + 4r + 4 = -1$$
$$(r+2)^2 = -1 \Rightarrow r + 2 = \pm i$$
$$r = -2 \pm i.$$

All solutions should be $C_1 y_1 + C_2 y_2$. "Basis" means $\{y_1, y_2\}$

Ans: $\boxed{\{e^{-2t}\cos t,\ e^{-2t}\sin t\}}$

(b) Procedure for undetermined coefficients.

Guess $y_p(t) = Ae^{-2t} + B\cos t + C\sin t$

(no need to multiply by t "overlap" means $Ae^{-2t}\cos t$ or $Be^{-2t}\sin t$)

Plug into $y'' + 4y' + 5y = e^{-2t} + \sin t$,

simplify; get a system of equations for A, B, C.

Solve to get A, B, C.

Answer: $C_1 e^{-2t}\cos t + C_2 e^{-2t}\sin t + Ae^{-2t} + B\cos t + C\sin t$

Not actual answer; need A, B, C for full credit.

Answer: $\boxed{C_1 e^{-2t}\cos t + C_2 e^{-2t}\sin t + e^{-2t} - \frac{1}{8}\cos t + \frac{1}{8}\sin t}$

$A = 1,\ B = -\frac{1}{8},\ C = \frac{1}{8}$

3. (25pts total) Consider the following system of linear equations, where a is a constant:

$$\begin{aligned} x_1 + x_2 - x_3 &= a \\ x_1 + ax_2 + 2x_3 &= a \\ 2x_1 + 2x_2 + ax_3 &= -4 \end{aligned}$$

Do the following in any order:

(a) Find all numbers a for which the system is consistent.

(b) Find all values of a for which the system has infinitely many solutions.

(c) Choose one of the values a for which the system has infinitely many solutions (there is at least one). For that value a, write the solution to the system in parametric vector form.

$$\left[\begin{array}{ccc|c} 1 & 1 & -1 & a \\ 1 & a & 2 & a \\ 2 & 2 & a & -4 \end{array}\right] \sim \left[\begin{array}{ccc|c} 1 & 1 & -1 & a \\ 0 & a-1 & 3 & 0 \\ 0 & 0 & a+2 & -4-2a \end{array}\right].$$

* If $a \neq 1$ & $a = -2$, there are pivots in every row, so will be solvable.

* If $a = 1$, matrix is $\left[\begin{array}{ccc|c} 1 & 1 & -1 & 1 \\ 0 & 0 & 3 & 0 \\ 0 & 0 & 3 & -6 \end{array}\right] \sim \left[\begin{array}{ccc|c} 1 & 1 & -1 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -6 \end{array}\right]$ inconsistent.

* If $a = -2$, matrix is $\left[\begin{array}{ccc|c} 1 & 1 & -1 & -2 \\ 0 & -3 & 3 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right] \sim \left[\begin{array}{ccc|c} 1 & 0 & 0 & -2 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right].$

No pivot in right column: consistent; no pivot in col 3 $\Rightarrow$ infinitely many solutions

(a) $\boxed{\text{All } a \text{ except } 1}$ (b) $\boxed{-2}$

(c) x_3 free, $x_1 = -2$, $x_2 = x_3$.

$$\boxed{\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} -2 \\ x_3 \\ x_3 \end{bmatrix} = x_3 \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} + \begin{bmatrix} -2 \\ 0 \\ 0 \end{bmatrix}}$$

4. (25pts) Find a singular value decomposition of the following matrix A:

$$A = \begin{bmatrix} -2 & 4 \\ 2 & -4 \\ 1 & -2 \end{bmatrix}$$

$A^TA = \begin{bmatrix} 9 & -18 \\ -18 & 36 \end{bmatrix}$. ← Work skipped

Eigenvalues of A^TA: 45 & 0

Tells you $\Sigma_1 = \begin{bmatrix} \sqrt{45} & 0 \\ 0 & \sqrt{0} \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} 3\sqrt{5} & 0 \\ 0 & 0 \\ 0 & 0 \end{bmatrix}$

Corresponding eigenvectors of A^TA: $\begin{bmatrix} -1 \\ 2 \end{bmatrix}$ & $\begin{bmatrix} 2 \\ 1 \end{bmatrix}$; normalize to get $V = \begin{bmatrix} -1/\sqrt{5} & 2/\sqrt{5} \\ 2/\sqrt{5} & 1/\sqrt{5} \end{bmatrix} = \frac{1}{\sqrt{5}}\begin{bmatrix} -1 & 2 \\ 2 & 1 \end{bmatrix}$

$\vec{u}_1$ = 1st column of U is $\frac{1}{3\sqrt{5}} A \begin{bmatrix} -1/\sqrt{5} \\ 2/\sqrt{5} \end{bmatrix} = \frac{1}{15}\begin{bmatrix} -2 & 4 \\ 2 & -4 \\ 1 & -2 \end{bmatrix}\begin{bmatrix} -1 \\ 2 \end{bmatrix} = \begin{bmatrix} 2/3 \\ -2/3 \\ -1/3 \end{bmatrix}$.

2nd & 3rd columns must be orthogonal to $\vec{u}_1$, so in $\mathrm{Nul}([2 \ -2 \ -1])$.

Basis for $\mathrm{Nul}([2 \ -2 \ -1])$ is $\left\{\begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ 2 \end{bmatrix}\right\}$. ← work skipped

Orthogonalize by Gram-Schmidt $\left\{\begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ -1 \\ 4 \end{bmatrix}\right\}$; normalize $\left\{\frac{1}{\sqrt{2}}\begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \frac{1}{\sqrt{18}}\begin{bmatrix} 1 \\ -1 \\ 4 \end{bmatrix}\right\}$.

$$\boxed{U = \begin{bmatrix} \frac{2}{3} & \frac{1}{\sqrt{2}} & \frac{1}{3\sqrt{2}} \\ -\frac{2}{3} & \frac{1}{\sqrt{2}} & -\frac{1}{3\sqrt{2}} \\ -\frac{1}{3} & 0 & \frac{4}{3\sqrt{2}} \end{bmatrix}}$$

Ans: If $U = \begin{bmatrix} \frac{2}{3} & \frac{1}{\sqrt{2}} & \frac{1}{3\sqrt{2}} \\ -\frac{2}{3} & \frac{1}{\sqrt{2}} & -\frac{1}{3\sqrt{2}} \\ -\frac{1}{3} & 0 & \frac{4}{3\sqrt{2}} \end{bmatrix}$, $\Sigma_1 = \begin{bmatrix} 3\sqrt{5} & 0 \\ 0 & 0 \\ 0 & 0 \end{bmatrix}$, $V = \frac{1}{\sqrt{5}}\begin{bmatrix} -1 & 2 \\ 2 & 1 \end{bmatrix}$

then $A = U\Sigma_1 V^T$, $U^T = U^{-1}$, and $V^T = V^{-1}$

There is only one Σ_1; there are 3 other Vs due to $\pm$s; and many choices for U.

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5. (25pts) Let:

$$A = \begin{bmatrix} 1 & 2 \\ 2 & 0 \\ 0 & 2 \\ 1 & 1 \end{bmatrix}, \qquad \mathbf{y} = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix},$$

and $W = \operatorname{Col} A$. Let $T\colon \mathbb{R}^4 \to \mathbb{R}^4$ be the orthogonal projection onto W.

(a) Find the matrix of T. (For partial credit, find $T(\mathbf{y})$.)

(b) Explain why $A^T(\mathbf{p} - T(\mathbf{p})) = \mathbf{0}$ for any $\mathbf{p} \in \mathbb{R}^4$. (The "$T$" in "$A^T$" is "transpose".)

5(a): Intentionally messiest problem computationally on this practice test...

If you get a very messy problem on final:

↳ Do other problems first!

↳ Organize work carefully; don't do work in head.

↳ If you can't finish, write out steps.

↳ Think about if there are multiple ways to solve problem, see which is best.

① ↳ $T(\vec{p})$ is $A\hat{x}$, where $\hat{x}$ is a solution to $A^TA\hat{x} = A^T\vec{p}$. That means $\hat{x} = (A^TA)^{-1}A^T\vec{p}$,

so $T(\vec{p}) = A\hat{x} = A[(A^TA)^{-1}A^T]\vec{p} = [A(A^TA)^{-1}A^T]\vec{p}$; answer is $A\underbrace{(A^TA)^{-1}}_{\text{do this first, by itself.}}A^T$

$A^TA = \begin{bmatrix} 6 & 3 \\ 3 & 9 \end{bmatrix}$. $(A^TA)^{-1} = \frac{1}{15}\begin{bmatrix} 3 & -1 \\ -1 & 2 \end{bmatrix}$. $A(A^TA)^{-1}A^T = \frac{1}{15}\begin{bmatrix} 1 & 2 \\ 2 & 0 \\ 0 & 2 \\ 1 & 1 \end{bmatrix}\begin{bmatrix} 3 & -1 \\ -1 & 2 \end{bmatrix}\begin{bmatrix} 1 & 2 & 0 & 1 \\ 2 & 0 & 2 & 1 \end{bmatrix} = \frac{1}{15}\begin{bmatrix} 7 & 2 & 6 & 4 \\ 2 & 12 & -4 & 4 \\ 6 & -4 & 8 & 2 \\ 4 & 4 & 2 & 3 \end{bmatrix}$

work skipped.

② ↳ Gram-Schmidt on Col A to get orthogonal basis $\{\vec{v}_1, \vec{v}_2\}$

Ans: $[\underbrace{T(\vec{e}_1)}_{\frac{\vec{e}_1\cdot\vec{v}_1}{\vec{v}_1\cdot\vec{v}_1}\vec{v}_1 + \frac{\vec{e}_1\cdot\vec{v}_2}{\vec{v}_2\cdot\vec{v}_2}\vec{v}_2} \quad T(\vec{e}_2) \quad T(\vec{e}_3) \quad T(\vec{e}_4)]$

$\vec{v}_1 = \begin{bmatrix} 1 \\ 2 \\ 0 \\ 1 \end{bmatrix}$, $\vec{v}_2 = \begin{bmatrix} 3 \\ -2 \\ 4 \\ 1 \end{bmatrix}$ work skipped.

$T(\vec{e}_1) = \frac{\vec{e}_1\cdot\vec{v}_1}{\vec{v}_1\cdot\vec{v}_1}\vec{v}_1 + \frac{\vec{e}_1\cdot\vec{v}_2}{\vec{v}_2\cdot\vec{v}_2}\vec{v}_2 = \frac{1}{6}\begin{bmatrix} 1 \\ 2 \\ 0 \\ 1 \end{bmatrix} + \frac{3}{30}\begin{bmatrix} 3 \\ -2 \\ 4 \\ 1 \end{bmatrix} = \begin{bmatrix} 7/15 \\ 2/15 \\ 6/15 \\ 4/15 \end{bmatrix}$ (repeat for other 3 columns)

UU^T not wrong, but very messy.

5(b): *Not as bad as (a); also don't need (a) for (b)

*Check if you can solve (b), even if you can't solve (a).

*[Or give answer to (b) as a formula in terms of answer to (a)]

$T(\vec{p})$ is $\hat{p}$. So $p - T(p) = p - \hat{p}$ is $\perp$ to Col A.
~~So~~ But $(\operatorname{Col} A)^\perp = \operatorname{Nul}(A^T)$, so $A^T(\vec{p} - T(\vec{p})) = \vec{0}$.

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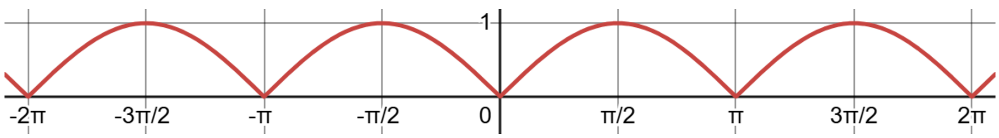
6. (25pts) Let $f(x) = \sin(x)$.

   (a) Find the Fourier sine series of $f(x)$ on $[0, \pi]$.

   (b) Show using Fourier series that for all x,

$$\frac{2}{\pi} - \frac{4}{\pi} \sum_{\substack{k \text{ even}, \\ k \geq 2}} \frac{1}{k^2 - 1} \cos(kx) = |\sin(x)|.$$

The graph of $|\sin(x)|$ is shown below:



The identities $\sin(a + b) = \sin(a)\cos(b) + \cos(a)\sin(b)$ and $\sin(a - b) = \sin(a)\cos(b) - \cos(a)\sin(b)$ may be helpful.

(a) $b_n = \frac{2}{\pi}\int_0^\pi \sin(x)\sin(nx)\,dx \rightsquigarrow \begin{cases} 1 & \text{if } n=1 \\ 0 & \text{if not.} \end{cases}$

So $1\cdot\sin(x) + 0\sin(2x) + 0\sin(3x) + \dots = \boxed{\sin x}$

Orthogonality: $\{\cos\frac{0\pi x}{L}, \cos\frac{1\pi x}{L}, \cos\frac{2\pi x}{L}, \dots, \sin\frac{1\pi x}{L}, \sin\frac{2\pi x}{L}, \dots\}$ is orthogonal under $\langle f, g\rangle = \int_{-L}^{L} f(x)g(x)\,dx$.

If $n \neq 0$, $L = \pi$ $\langle \sin x, \sin(nx)\rangle = 0$

$\int_{-\pi}^{\pi} \sin x \sin(nx)\,dx = 2\cdot\int_0^\pi \sin x \sin(nx)\,dx$

Keep going, using $\sin(a)\sin(b) = \frac{1}{2}\cos(a-b) - \frac{1}{2}\cos(a+b)$.

Not wrong, but way too long.

Fourier sine series is best approximation to $f(x)$ as an infinite linear combination of $\sin(\frac{\pi x}{L}), \sin(\frac{2\pi x}{L}), \dots$
$\sin x$ already is a linear combination, so answer is $\boxed{\sin x} = 1\sin x + 0\sin 2x + 0\sin 3x + \dots$

(b) Find Fourier cosine series of $f(x)$.

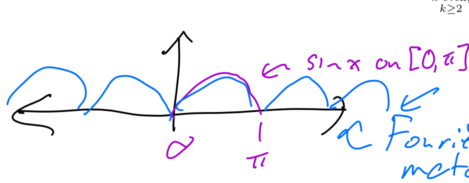
Use $\sin(a + b) = \sin(a)\cos(b) + \cos(a)\sin(b)$ and $\sin(a - b) = \sin(a)\cos(b) - \cos(a)\sin(b)$ to get $\sin(a)\cos(b) = \frac{1}{2}\sin(a+b) + \frac{1}{2}\sin(a-b)$

Should get $a_n = \frac{2}{\pi}\int_0^\pi \sin x\cos(nx)\,dx = \frac{1}{\pi}\int_0^\pi [\sin((\underbrace{\cancel{n+1}}_{1+n})x) + \sin((\underbrace{\cancel{n-1}}_{1-n})x)]\,dx$

if $n=1$: $\frac{1}{\pi}\int_0^\pi \sin(2x)\,dx = \frac{1}{2\pi}\left[-\cos 2x\right]_{x=0}^{\pi} = 0.$

if $n \neq 1$: (work skipped) $= \frac{2[(-1)^n+1]}{\pi(1-n^2)} = \begin{cases} -\frac{4}{\pi(n^2-1)} & \text{if } n \text{ even} \\ 0 & \text{if } n \text{ odd.} \end{cases}$

So Fourier cosine series of f is $\frac{2}{\pi} - \frac{4}{\pi} \sum_{\substack{k \text{ even}, \\ k \geq 2}} \frac{1}{k^2 - 1} \cos(kx)$. But also, can get Fourier cosine series of f as follows:



2π-periodic extension of even extension of f. No jumps, so blue is Fourier cosine series of f.

Fourier cosine matches $|\sin x|$.

So both $\frac{2}{\pi} - \frac{4}{\pi} \sum_{\substack{k \text{ even}, \\ k \geq 2}} \frac{1}{k^2 - 1} \cos(kx)$ and $|\sin x|$ are the Fourier cosine series of f.

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7A (25pts) **For Prof. Serganova's lecture only; Prof. Sharma's students should solve 7B on the next page.** [This question is fair for Prof. Sharma.]

Let $\mathbf{a}$ and $\mathbf{b}$ be any vectors in $\mathbb{R}^n$ with $\mathbf{a}\cdot\mathbf{b}\neq 0$. Let $T\colon \mathbb{R}^n \to \mathbb{R}^n$ be the function

$$T(\mathbf{x}) = \frac{\mathbf{a}\cdot\mathbf{x}}{\mathbf{a}\cdot\mathbf{b}}\mathbf{b}$$

(a) Show that T is a linear transformation.

(b) Describe which vectors are in $\operatorname{Ker} T$.

(c) Find a basis for $\operatorname{Ran} T$. Range of T. (All outputs of T.)

(d) Let A be the matrix of T. Explain why A is diagonalizable, and find a matrix D for which there is an invertible matrix P with $A = PDP^{-1}$.

(a) Check $T(\vec{x}+\vec{y}) = T(\vec{x}) + T(\vec{y})$ & $T(c\vec{x}) = cT(\vec{x})$

(b) Since $\vec{a}\cdot\vec{b}\neq 0$, $\vec{b}$ can't be 0, so $T(\vec{x})=\vec{0}$ means $\frac{\vec{a}\cdot\vec{x}}{\vec{a}\cdot\vec{b}}=0$, so $\vec{x}$ is $\perp$ to $\vec{a}$. So $\operatorname{Ker} T = \boxed{\{\vec{a}\}^\perp}$

(c) T only outputs scalar multiples of $\vec{b}$, and $\vec{b}$ is an output of T ($T(\vec{b}) = \cancel{\frac{\vec{a}\cdot\vec{b}}{\vec{a}\cdot\vec{b}}}\vec{b} = \vec{b}$), so $\operatorname{Ran}(T) = \operatorname{Span}(\{\vec{b}\})$, and $\boxed{\{\vec{b}\}}$ is a basis for Ran T.

(d) * Part b says: each vector $\vec{x}$ in $\{\vec{a}\}^\perp$ satisfies $T(\vec{x})=\vec{0}$ so $A\vec{x}=\vec{0}$. There are $n-1$ linearly independent vectors in $\{\vec{a}\}^\perp$, and so this gives $n-1$ linearly independent eigenvectors of A for 0.

* As in part (c), $T(\vec{b})=\vec{b}$. So $\vec{b}$ is an eigenvector of A with eigenvalue 1.

* The two together give n linearly independent eigenvectors of A. So A is diagonalizable,

$$\boxed{D = \begin{bmatrix} 0 & & & \text{\Large 0} \\ & 0 & & \\ & & \ddots & \\ & & & 0 & \\ \text{\Large 0} & & & & 1 \end{bmatrix}}$$

($n-1$ "0"s, one "1" on diagonal.)

4:47

7B (25pts) **For Prof. Sharma's lecture only; Prof. Serganova's students should solve 7A on the previous page.** [This question is fair for Prof. Serganova.]

Let $V = M_{2\times 2}$, the vector space of all 2×2 matrices with real coefficients. Let H be the set of all symmetric 2×2 matrices.

(a) Show that H is a subspace of V.

(b) Find a basis for H and the dimension of H.

(a) To show H is a subspace of V, need to check:

- Zero vector of V in H: Zero vector of V is $\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$ is symmetric
- Closed under $+$ Could do $\begin{bmatrix} a & b \\ c & d \end{bmatrix} + \begin{bmatrix} e & f \\ g & h \end{bmatrix}$

  Let A & B be in H. So $A^T = A$ & $B^T = B$. That means
  $(A+B)^T = A^T + B^T = A+B$, So $A+B$ is symmetric, so in H.
- Closed under $\cdot$ You try.

(b) Let $\begin{bmatrix} a & b \\ c & d \end{bmatrix}$ be any matrix in V.

To be in H, need $b=c$, so

$$\begin{bmatrix} a & b \\ b & d \end{bmatrix} = a\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} + b\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} + d\begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$$

So $\left\{\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}\right\}$ spans H.

Linear independence:

If $a\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} + b\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} + d\begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$

then $\begin{bmatrix} a & b \\ b & d \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$ so $a=0, b=0, d=0$.

$\boxed{\left\{\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}\right\}}$ is a basis, H has dimension $\boxed{3}$.

8A (25pts) **For Prof. Serganova's lecture only; Prof. Sharma's students should solve 7B on the next page.** [Parts (a) and (b) can be solved using content Prof. Sharma covered, but not part (c)].]

(a) Let $\mathbf{A}(t)$ be any $n \times n$ matrix whose entries are functions that are continuous for $a < t < b$. Let $\mathbf{X}(t)$ be a fundamental matrix for the system of ODEs:

$$\mathbf{x}'(t) = \mathbf{A}(t)\mathbf{x}(t) \qquad a < t < b.$$

Show that $\mathbf{X}'(t) = \mathbf{A}(t)\mathbf{X}(t)$.

(b) Using part (a), find a 2×2 matrix $\mathbf{A}(t)$ of functions that are continuous on $(0, 2)$ for which $\begin{bmatrix} e^{5t} \\ 2e^{5t} \end{bmatrix}$ and $\begin{bmatrix} t \\ t^2 \end{bmatrix}$ are both solutions to $\mathbf{x}'(t) = \mathbf{A}(t)\mathbf{x}(t)$.

(c) Explain why there is no 2×2 matrix $\mathbf{A}(t)$ of functions that are continuous on $(-\infty, \infty)$ for which $\begin{bmatrix} e^{5t} \\ 2e^{5t} \end{bmatrix}$ and $\begin{bmatrix} t \\ t^2 \end{bmatrix}$ are both solutions to $\mathbf{x}'(t) = \mathbf{A}(t)\mathbf{x}(t)$.

(a) Since $X(t)$ is a fundamental matrix of $\vec{x}'(t) = A(t)\vec{x}(t)$, each column of $X(t)$ is a solution to $\vec{x}' = A(t)\vec{x}$. So if $\vec{x}_1(t), \dots, \vec{x}_n(t)$ are the columns of $X(t)$, then:

$$\begin{aligned} A(t)X(t) &= A(t)\left[\vec{x}_1(t) \;\cdots\; \vec{x}_n(t)\right] \\ &= \left[A(t)\vec{x}_1(t) \;\cdots\; A(t)\vec{x}_n(t)\right] \\ &= \left[\vec{x}_1'(t) \;\cdots\; \vec{x}_n'(t)\right] \\ &= \frac{d}{dt}\left[\vec{x}_1(t) \;\cdots\; \vec{x}_n(t)\right] \\ &= \frac{d}{dt}X(t) \checkmark \end{aligned}$$

(b) Since $\begin{bmatrix} e^{5t} \\ 2e^{5t} \end{bmatrix}$ & $\begin{bmatrix} t \\ t^2 \end{bmatrix}$ are both solutions to $\vec{x}'(t) = A(t)\vec{x}(t)$, $X(t) = \begin{bmatrix} e^{5t} & t \\ 2e^{5t} & t^2 \end{bmatrix}$ is a fundamental matrix of $\vec{x}' = A(t)\vec{x}$. So by (a)

$$X'(t) = A(t)X(t),$$

so $A(t) = [X'(t)][X(t)]^{-1}$ where $X(t) = \begin{bmatrix} e^{5t} & t \\ 2e^{5t} & t^2 \end{bmatrix}$

$= $ (work skipped) $\boxed{\begin{bmatrix} \frac{5t^2-2}{t(t-2)} & \frac{-5t+1}{t(t-2)} \\ \frac{2(5t-2)}{t-2} & \frac{-8}{t-2} \end{bmatrix}}$

(c) Key fact about Wronskians: If $\vec{x}_1(t), \dots, \vec{x}_n(t)$ are all solutions to $\vec{x}' = A(t)\vec{x}$ for $a < t < b$ and $A(t)$ is continuous on $a < t < b$, then $W[\vec{x}_1, \dots, \vec{x}_n](t)$ is 0 everywhere or 0 nowhere on (a,b).
But Wronskian of $\begin{bmatrix} e^{5t} \\ 2e^{5t} \end{bmatrix}$ and $\begin{bmatrix} t \\ t^2 \end{bmatrix}$ is $\det\begin{bmatrix} e^{5t} & t \\ 2e^{5t} & t^2 \end{bmatrix} = e^{5t}t(t-2)$ which is both 0 & not 0 on $(-\infty,\infty)$. So no such $A(t)$ for $-\infty < t < \infty$.

Spring 2026 Topic Reviews for Math 54

Faculty: Profs. Arun Sharma and Vera Serganova
Facilitator: Jason Ferguson, jason.ferguson@berkeley.edu
Time and Location: Tu 3:30-5 on Zoom and in 201A Student Center, Th 3:30-5 on Zoom and 6 Evans
Jason's Office Hours: As of 3/16, M 12-1, TuTh 2-3 See here for the most up-to-date Drop-In Schedule

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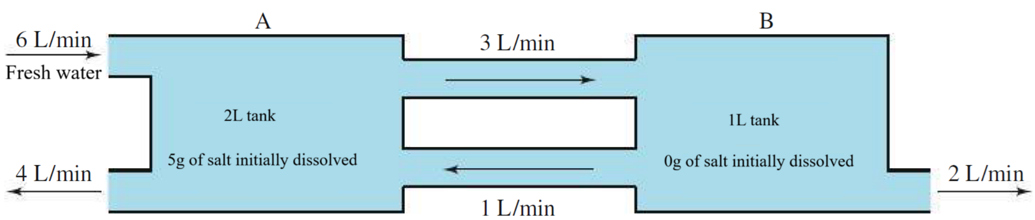
8B (25pts) **For Prof. Sharma's lecture only; Prof. Serganova's students should solve 8A on the previous page.**

In the following setup for a mixing problem:



find a formula for the concentration of salt in Tank A after t minutes, in g/L.

let $x_1(t)$ = # of g of salt in Tank A after t mins

$x_2(t)$ = '' for tank B,

rate salt enters/leaves in $\frac{g}{min}$

flowing out

flowing in.

$$x_1'(t) = -\frac{7}{2}x_1(t) + x_2(t) \qquad x_1(0) = 5$$

$$x_2'(t) = \frac{3}{2}x_1(t) - 3x_2(t) \qquad x_2(0) = 0,$$

let $\vec{x}(t) = \begin{bmatrix} x_1(t) \\ x_2(t) \end{bmatrix}$, $A = \begin{bmatrix} -\frac{7}{2} & 1 \\ \frac{3}{2} & -3 \end{bmatrix}$, so $\vec{x}' = A\vec{x}$, $\vec{x}(0) = \begin{bmatrix} 5 \\ 0 \end{bmatrix}$.

A's eigenvalues are $-\frac{9}{2}$ & -2 with eigenvectors $\begin{bmatrix} -1 \\ 1 \end{bmatrix}$ & $\begin{bmatrix} 2 \\ 3 \end{bmatrix}$.

$\vec{x}(t) = C_1 e^{-\frac{9}{2}t}\begin{bmatrix} -1 \\ 1 \end{bmatrix} + C_2 e^{-2t}\begin{bmatrix} 2 \\ 3 \end{bmatrix}$. $\vec{x}(0) = \begin{bmatrix} 5 \\ 0 \end{bmatrix}$ means $C_1 = -3$, $C_2 = 1$,

Problem wants $\frac{x_1(t)}{2} = \boxed{\frac{3}{2}e^{-\frac{9}{2}t} + e^{-2t}}$

Concentration

9 (5pts each = 25 points in all) For each of the following, fill in the bubble to decide if the statement is true or false. Correct answers will receive 5 points; answers left blank will receive 2.5 points; and incorrect answers will receive 0 points.

Explanations are not needed and will not be looked at, but writing them out may still help you in your own studying or deciding the correct answer.

(a) Eigenvectors for distinct eigenvalues of a real symmetric matrix are orthogonal.

● TRUE ○ FALSE

Spectral theorem.

(b) If A and B are similar matrices, then A and B have the same characteristic polynomial.

● TRUE ○ FALSE

"A & B similar" means $A = P^{-1}BP$ for some invertible P.

$\det(A-\lambda I) = \det(P^{-1}BP - \lambda I) = \det(P^{-1}BP - \lambda P^{-1}IP)$

$= \det(P^{-1}(B-\lambda I)P) = \det(P^{-1})\det(B-\lambda I)\det(P) = \det(B-\lambda I)$.

✓

(c) If A is diagonalizable, then A^T is also diagonalizable, and A^T has the same eigenvalues with the same multiplicity as A.

● TRUE ○ FALSE

If $A = PDP^{-1}$ then $A^T = (P^{-1})^T D^T P^T = \underbrace{(P^T)^{-1} D P^T}$,

shows A^T is diagonalizable with same D.

(d) For any linear transformation $T\colon V \to W$ and vector spaces V and W, If $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$ is linearly dependent, then $\{T(\mathbf{v}_1), T(\mathbf{v}_2), T(\mathbf{v}_3)\}$ is linearly dependent.

● TRUE ○ FALSE

There are numbers c_1, c_2, c_3, not all 0 with

$c_1\vec{v}_1 + c_2\vec{v}_2 + c_3\vec{v}_3 = \vec{0}$

Then $c_1 T(\vec{v}_1) + c_2 T(\vec{v}_2) + c_3 T(\vec{v}_3) = \vec{0}$.

(e) For any linear transformation $T\colon V \to W$ and vector spaces V and W, If $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$ is linearly independent, then $\{T(\mathbf{v}_1), T(\mathbf{v}_2), T(\mathbf{v}_3)\}$ is linearly independent.

○ TRUE ● FALSE

If $T(\vec{x}) = \vec{0}$ for all $\vec{x}$, then $\{T(\vec{v}_1), T(\vec{v}_2), T(\vec{v}_3)\}$ is dependent.

(Need T to be one-to-one.)

10 (5pts each = 25 points in all) For each of the following, fill in the bubble to decide if the statement is true or false. Correct answers will receive 5 points; answers left blank will receive 2.5 points; and incorrect answers will receive 0 points.

Explanations are not needed and will not be looked at, but writing them out may still help you in your own studying or deciding the correct answer.

(a) For any $n \times n$ matrix A, the formula $\langle \mathbf{x}, \mathbf{y} \rangle = (A\mathbf{x}) \cdot (A\mathbf{y})$ is an inner product on $\mathbb{R}^n$.

○ TRUE ● FALSE

Need A to have linearly independent columns. Otherwise $A\vec{x} = \vec{0}$ has a nontrivial solution, and for that $\vec{x}$,

$\langle \vec{x}, \vec{x} \rangle = (A\vec{x}) \cdot (A\vec{x}) = \vec{0} \cdot \vec{0} = 0$

even though $\vec{x} \neq \vec{0}$.

(b) If A is an orthogonal matrix, then the linear map $T(\mathbf{x}) = A\mathbf{x}$ is one-to-one and onto.

● TRUE ○ FALSE

A is invertible. (Its inverse is A^T.)

(c) All eigenvalues of any non-singular matrix are real numbers.

○ TRUE ● FALSE

$\begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$ has eigenvalues $\pm i$.

(Correct statement is: nonzero numbers.)

(d) For any 5×6 matrix A, if $\operatorname{Col} A \neq \mathbb{R}^5$, then $\operatorname{Nul} A$ is at least 2-dimensional

● TRUE ○ FALSE

$\dim \operatorname{Nul} A = 6 - \dim(\operatorname{Col} A)$.

Since $\operatorname{Col} A \neq \mathbb{R}^5$, $\dim \operatorname{Col} A \leq 4$, so $\dim \operatorname{Nul} A \geq 2$.

(e) There is only one 4×3 matrix with rank 3 that is also in reduced echelon form.

● TRUE ○ FALSE

It's $\rightarrow \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}$.

Rank 3 means 3 pivot columns. Everything above & below pivots are 0. Pivots must be where they're drawn.

11 (5pts each = 25 points in all) For each of the following, fill in the bubble to decide if the statement is true or false. Correct answers will receive 5 points; answers left blank will receive 2.5 points; and incorrect answers will receive 0 points.

Explanations are not needed and will not be looked at, but writing them out may still help you in your own studying or deciding the correct answer.

(a) For any $n \times n$ non-diagonalizable real matrix A, if λ is a real eigenvalue of A with eigenvector $\mathbf{u}$, then $\mathbf{x}(t) = e^{\lambda t}\mathbf{u}$ is a solution to $\mathbf{x}'(t) = A\mathbf{x}(t)$.

● TRUE ○ FALSE

If $\vec{x}(t) = e^{\lambda t}\vec{u}$, then $\vec{x}\,'(t)$ & $A\vec{x}(t)$ are both $\lambda e^{\lambda t}\vec{u}$.

(b) For any $n \times n$ real matrix A, if if every eigenvalue of A is a complex (non-real) number, then $\mathbf{x}'(t) = A\mathbf{x}(t)$ has no real solutions besides $\mathbf{x}(t) = \mathbf{0}$.

○ TRUE ● FALSE

$\vec{x}\,'(t) = A\vec{x}(t)$ has n linearly independent real solutions (will have sines & cosines if all eigenvalues are non-real.)

(c) The dimensions of the row space and of the column space of any matrix A are always the same, whether or not A is square.

● TRUE ○ FALSE

Both are # of pivots.

(d) If $f(x)$ is any twice-differentiable odd function, then for any $L > 0$, the Fourier sine series of $f(x)$ on $[0, L]$ is the Fourier series of $f(x)$ on $[-L, L]$.

● TRUE ○ FALSE

$$\frac{1}{L}\int_{-L}^{L} \underbrace{f(x)\sin\left(\frac{n\pi x}{L}\right)}_{\text{even function}} dx = \frac{2}{L}\int_0^L f(x)\sin\left(\frac{n\pi x}{L}\right) dx$$

So b_n's are same; a_n for Fourier series is 0

(e) If $f(x)$ is any twice-differentiable odd function, then for any $L > 0$, the Fourier cosine series of $f(x)$ on $[0, L]$ is 0.

○ TRUE ● FALSE

See problem 6b for counterexample.

(Correct statement is: $\cancel{b_n}$ a_n's in Fourier series of f on $[-L,L]$ are all 0.)

12 (15 points) Find the general solution to the following system of differential equations:

$$u'(t) = -u(t) - v(t) - v'(t),$$
$$v''(t) = u(t) - 4v(t) - 2v'(t).$$

Hint: One of the roots of the cubic is -1.

Method 1: Let $\vec{x}(t) = \begin{bmatrix} u(t) \\ v(t) \\ v'(t) \end{bmatrix}$, $A = \begin{bmatrix} -1 & -1 & -1 \\ 0 & 0 & 1 \\ 1 & -4 & -2 \end{bmatrix}$.

Then need to solve $\vec{x}'(t) = A\vec{x}(t)$.

synthetic division.

Char. poly: $-\lambda^3 - 3\lambda^2 - 7\lambda - 5 = -(\lambda+1)(\lambda^2+2\lambda+5)$

(work skipped)

A's eigenvalues are -1 & $-1+2i$, & $-1-2i$

Eigenvectors $\begin{bmatrix} -3 \\ -1 \\ 1 \end{bmatrix}$ & $\begin{bmatrix} -1 \\ 1 \\ -1+2i \end{bmatrix}$

Solutions to $\vec{x}' = A\vec{x}$ are

$$\vec{x}(t) = C_1 e^{-t}\begin{bmatrix} -3 \\ -1 \\ 1 \end{bmatrix} + C_2 e^{-t}\left((\cos 2t)\begin{bmatrix} -1 \\ 1 \\ -1 \end{bmatrix} - (\sin 2t)\begin{bmatrix} 0 \\ 0 \\ 2 \end{bmatrix}\right) + C_3 e^{-t}\left((\sin 2t)\begin{bmatrix} -1 \\ 1 \\ -1 \end{bmatrix} + (\cos 2t)\begin{bmatrix} 0 \\ 0 \\ 2 \end{bmatrix}\right)$$

So
$$\boxed{\begin{aligned} u(t) &= -3C_1 e^{-t} - C_2 e^{-t}\cos 2t - C_3 e^{-t}\sin 2t \\ v(t) &= -C_1 e^{-t} + C_2 e^{-t}\cos 2t + C_3 e^{-t}\sin 2t. \end{aligned}}$$

Method 2: Solve $v''(t) = u(t) - 4v(t) - 2v'(t)$. for $u(t)$, get $u(t) = v''(t) + 2v'(t) + 4v(t)$

Plug into $u'(t) = -u(t) - v(t) - v'(t)$, get $v''' + 2v'' + 4v' = -(v'' + 2v' + 4v) - v - v'$

$v''' + 3v'' + 7v' + 5v = 0.$

Roots of aux. polynomial: -1, $-1 \pm 2i$. (work skipped)

$$\boxed{v(t) = C_1 e^{-t} + C_2 e^{-t}\cos 2t + C_3 e^{-t}\sin 2t.}$$

Plug into $u(t) = v''(t) + 2v'(t) + 4v(t)$; get (work skipped)

$$\boxed{u(t) = 3C_1 e^{-t} - C_2 e^{-t}\cos 2t - C_3 e^{-t}\sin 2t}$$

Scratch work